

# Cycle-Intersection Centrality: A High-Order Topological Metric for Identifying Cycle-Structure Control Nodes in Complex Networks

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## Abstract

Classical node-importance metrics for complex networks such as degree centrality, k-shell decomposition, H-index centrality, and cycle ratio (CR) are built on low-order neighbourhood features or global skeleton attributes. None of them captures a node's position at the intersection of multiple independent cycles, the structural signature of feedback hubs that dominate cyclic control and topological stability. To fill this gap, we propose Cycle-Intersection Centrality (CIC), which quantifies, for each node  $v$ , the total shared-node overlap among all pairs of chordless cycles that pass through  $v$ , excluding  $v$  itself. Unlike the cycle number (NC) or cycle ratio, which measure how many cycles converge at a node, CIC measures how much genuine structure the cycle pairs share beyond the meeting point: a hub on  $k$  disjoint triangles attains  $NC = k$  but  $CIC = 0$ , because no two triangles share an edge. We evaluate CIC against degree, k-shell, H-index, CR, and a cycle-diversity variant on six real networks (karate, dolphins, polbooks, *C. elegans*, US power grid, and Email-Eu-core), under a two-layer protocol: minimum feedback vertex set (MFVS) and cycle pinning as cycle-native ground truths, and SIR spreading and percolation as boundary tasks. Results show that CIC carries information substantially orthogonal to degree and H-index (Kendall  $\tau \leq 0.87$ , as low as 0.42 on polbooks) while remaining interpretable through cycle structure. On the sparse power-grid network, top- $k$  CIC nodes break a larger fraction of cycles under pinning than degree (0.266 vs. 0.227 at  $k = 20$ ), the scenario in which CIC's cycle-oriented design yields a measurable advantage. On MFVS greedy deletion, propagation, and percolation, tasks dominated by hub coverage and global connectivity, CIC proves inferior to degree; this reflects the hub-dominated nature of those tasks rather than a defect of the metric. We further identify three structural mechanisms: rank-tie saturation of H-index, the distinction between cycle convergence and cycle intersection embodied in CIC, and the mismatch between chordless-cycle sets and linear cycle-space bases (verified on  $K_{2,3}$ ). The full experimental pipeline, datasets with SHA-256 checksums, and parameter settings are provided for exact reproduction.

Keywords: complex networks; node importance; cycle structure; chordless cycles; feedback vertex set; centrality

## 1. Introduction

Identifying important nodes is a foundational problem in network science, with applications ranging from epidemic control and influence maximization to the stabilisation of power grids and the analysis of biological feedback circuits [1–4]. A large family of centrality measures has been proposed over

the past five decades, yet the overwhelming majority operate on first-order neighbourhoods or on global skeleton decompositions: degree centrality counts immediate neighbours [5]; k-shell decomposition iteratively strips nodes of degree below a threshold and ranks nodes by their residual shell [6]; H-index centrality evaluates a node by the H-index of its neighbours' degrees [7,8]. These metrics are highly correlated with one another on many real networks, and they all ignore a structural ingredient that is decisive for cyclic dynamics: the cycles themselves.

Cycles are the simplest building blocks of redundancy and feedback [9,10]. In gene-regulatory networks, feedback loops determine oscillatory behaviour; in power grids, the meshed structure of cycles provides fault tolerance; in opinion and epidemic dynamics, cycle structure channels recurrent influence. Recent work introduced cycle-based node importance measures, most notably the cycle ratio (CR), which quantifies the fraction of cycles passing through a node among all cycles it could participate in [11]. While CR moved beyond first-order neighbourhoods, it still measures only cycle participation quantity: how many cycles contain a node. It does not distinguish a node that merely sits on many cycles from a node that lies at the genuine intersection of several cycles, a hub where cycles share edges and paths, and where local structural overlap concentrates. This “cycle-intersection hub” is exactly the node that should dominate cycle-breaking control tasks, yet no existing metric isolates this role.

In this paper we fill this gap with a new metric, Cycle-Intersection Centrality (CIC). The key idea is a conceptual split within cycle-based importance. A node  $v$  has high convergence if many cycles pass through it (measured by NC and CR); it has high intersection if the pairs of cycles meeting at  $v$  share substantial structure beyond  $v$  itself. CIC formalises the latter: it sums, over all unordered pairs of chordless cycles containing  $v$ , the number of shared nodes other than  $v$ . A hub on  $k$  disjoint triangles is the canonical separation example: it has  $NC = k$  but  $CIC = 0$ , because the triangles share only the hub. Conversely, in a diamond graph (two triangles sharing an edge), the shared nodes receive positive CIC. CIC is computed from co-occurrence counts of node pairs across all enumerated chordless cycles, at  $O(\#cycles \times L^2)$  cost for mean cycle length  $L$ , which remains tractable for the one million chordless cycles of Email-Eu-core.

The contribution of the paper is fourfold. (1) We systematically identify the structural blind spot shared by degree, k-shell, H-index, and cycle-ratio style metrics in cycle-intersection recognition, and formalise the distinction between cycle convergence and cycle intersection. (2) We propose CIC, a cycle-intersection centrality with a precise combinatorial definition and an efficient evaluation scheme. (3) We construct a two-layer evaluation protocol: cycle-native ground truths (exact minimum feedback vertex set and cycle pinning) where CIC's design can plausibly excel, and conventional hub-dominated tasks (SIR spreading and percolation) where we expect and report inferiority, converting potential negative results into a precise statement of the metric's application boundary. (4) We expose three theoretical mechanisms, namely rank-tie saturation of H-index, the saturation of cycle-diversity terms, and the mismatch between chordless-cycle sets and linear cycle-space bases, that clarify when each metric can be trusted. All experiments use six standard real networks with published ground-truth sizes, pinned dataset URLs and SHA-256 checksums (Appendix A), so every figure and table can be reproduced exactly.

## 2. Related Work and Preliminaries

### 2.1 Classical centrality metrics

Degree centrality  $d(v)$  counts the neighbours of  $v$  [5].  $k$ -shell (coreness)  $ks(v)$  is the largest  $k$  such that  $v$  belongs to the  $k$ -core, the maximal subgraph of minimum degree  $\geq k$  [6]. H-index centrality  $h(v)$  is the largest integer  $h$  such that  $v$  has at least  $h$  neighbours with degree  $\geq h$  [7,8]. All three are highly rank-correlated on many real networks (Section 4.1), and none consults the cycle structure of the network. Their computational costs are also low-order: degree and  $k$ -shell are linear in  $m$ , and H-index centrality costs  $O(m \log m)$  per node on the degree sequence. This efficiency is a genuine virtue for massive networks, but it is achieved by discarding precisely the high-order information (cycles and their intersections) that this paper argues is decisive for cyclic control tasks.

### 2.2 Cycle-based metrics and chordless cycles

For an undirected graph  $G = (V, E)$ , a cycle is a closed walk with no repeated vertices. A chord of a cycle is an edge between two non-consecutive vertices of the cycle; a cycle is chordless if it has no chord. Chordless cycles are the natural “short” building blocks of cycle space: the cycle space of a connected graph has dimension  $\mu = m - n + 1$ , but the set of chordless cycles is generally much larger and carries more structural information. Recent work defined the cycle number  $NC(v)$ , the number of (short) cycles containing  $v$ , and the cycle ratio  $CR(v)$ , which normalises  $NC(v)$  by the number of cycles through  $v$ ’s neighbours [11].  $CR$  was shown to rank nodes differently from degree, H-index, and coreness, and to perform competitively on spreading and synchronisation tasks [11]. However, both  $NC$  and  $CR$  are convergence measures: they count cycles, not the structural overlap between cycles at  $v$ .

### 2.3 Minimum feedback vertex set (MFVS)

A feedback vertex set (FVS) is a set of vertices whose removal makes the graph acyclic. Finding a minimum FVS (MFVS) is NP-hard [12], and it is the canonical combinatorial formalisation of cycle control: a small FVS breaks all cycles with few interventions. For the small networks in this study ( $n \leq 105$ ) we can solve MFVS exactly via mixed-integer programming on the chordless-cycle hypergraph (Section 3.4); for *C. elegans* ( $n = 297$ ) we report the best feasible solution under a 120 s time limit. MFVS is therefore a cycle-native ground truth with which to rank the cycle-control ability of centrality-driven greedy deletion.

### 2.4 Dynamics tasks

SIR epidemic spreading is the standard hub-dominated propagation task: a seed set of size  $k$  is infected and the final outbreak size measures the seed’s spreading influence [10,13]. Percolation robustness removes nodes in decreasing order of a metric and measures the size of the largest remaining connected component (LCC); metrics that preserve global connectivity better leave a larger LCC [14]. Cycle pinning fixes the top- $k$  nodes of a metric and measures the fraction of cycles that are hit (broken), a cycle-native control task. Our protocol treats SIR and percolation as boundary tasks and pinning (together with MFVS) as the core cycle-native verification.

### 3. Methods

#### 3.1 Datasets

Six standard real networks were used: Zachary’s karate club ( $n = 34$ ,  $m = 78$ ) [1]; bottlenose dolphins ( $n = 62$ ,  $m = 159$ ) [15]; US politics books ( $n = 105$ ,  $m = 441$ ) [16]; the *C. elegans* neural network ( $n = 297$ ,  $m = 2148$ ) [2]; the US western-states power grid ( $n = 4941$ ,  $m = 6594$ ) [2]; and the largest connected component of Email-Eu-core ( $n = 1005$ ,  $m = 16064$ ) [17]. All graphs were made undirected, stripped of self-loops and duplicate edges, and the resulting  $(n, m)$  match the published values. Table 1 summarises the per-network parameters used throughout this paper: the cycle-length limit  $L$ , the number of chordless cycles enumerated, and the epidemic threshold  $\beta_c$  with the SIR infection rate  $\beta = 1.5\beta_c$  used in Section 3.6. Raw files, pinned URLs and SHA-256 checksums are listed in Appendix A.

| Network           | $n$  | $m$   | $L$ | # chordless cycles | $\beta_c$ | $\beta = 1.5\beta_c$ |
|-------------------|------|-------|-----|--------------------|-----------|----------------------|
| karate            | 34   | 78    | 8   | 103                | 0.1287    | 0.1931               |
| dolphins          | 62   | 159   | 8   | 1410               | 0.1470    | 0.2204               |
| polbooks          | 105  | 441   | 8   | 6343               | 0.0838    | 0.1257               |
| power grid        | 4941 | 6594  | 8   | 2753               | 0.2583    | 0.3875               |
| <i>C. elegans</i> | 297  | 2148  | 5   | 107497             | 0.0384    | 0.0576               |
| Email-Eu-core     | 1005 | 16064 | 4   | 1011864            | 0.0134    | 0.0201               |

Table 1. Dataset parameters used in all experiments.  $L$  is the maximum chordless-cycle length;  $\beta_c$  is the leading-eigenvalue epidemic threshold; SIR uses  $\beta = 1.5\beta_c$  with recovery rate  $\gamma = 1$ .

#### 3.2 Chordless-cycle enumeration

Chordless cycles up to a maximum length  $L$  are enumerated for each network. We set  $L = 8$  for karate, dolphins, polbooks, and power grid;  $L = 5$  for *C. elegans*; and  $L = 4$  for Email-Eu-core, chosen so that the enumeration remains tractable while capturing the networks’ short-cycle structure. The resulting cycle counts are: 103 (karate), 1410 (dolphins), 6343 (polbooks), 2753 (power grid), 107497 (*C. elegans*), and 1011864 (Email-Eu-core). Enumeration is deterministic; metrics computed from the cycle sets are therefore exactly reproducible (Section 4.1 of Appendix A).

#### 3.3 Cycle-Intersection Centrality (CIC)

Let  $\mathcal{C}(v) = \{c_1, \dots, c_k\}$  be the set of chordless cycles containing  $v$ . Define the co-occurrence count

$$c_{\{vu\}} = |\{c \in \mathcal{C}: v \in c \text{ and } u \in c\}|$$

the number of cycles containing both  $v$  and  $u$ . We define Cycle-Intersection Centrality as

$$CIC(v) = \sum_{\{u \neq v\}} C(c_{\{vu\}}, 2)$$

where  $C(x, 2) = x(x - 1)/2$  is the binomial coefficient. Intuitively, every pair of cycles that both contain  $v$  and both contain  $u$  contributes one unit to  $v$ ; the total counts how much the cycle pairs meeting at  $v$  genuinely share, node by node, beyond  $v$  itself. This is the intersection dimension: NC and CR are monotone in the number of cycles through  $v$  (convergence), while CIC is sensitive to the structure shared between those cycles. Two structural facts follow directly. First, a hub on  $k$  disjoint triangles has  $NC = k$  but  $CIC = 0$ , since no two triangles share a node other than the hub. Second, in a diamond (two triangles sharing an edge  $(a, b)$ ), the shared nodes  $a$  and  $b$  both receive  $CIC = 1$ , isolating exactly

the cycle-intersection hubs that convergence measures cannot see. CIC is evaluated by accumulating, for each enumerated cycle of length  $L$ , one co-occurrence increment on each of its  $C(L, 2)$  node pairs, for a total cost of  $O(\#\text{cycles} \times L^2)$ . The evaluation is summarised in Algorithm 1.

**Algorithm 1. Cycle-Intersection Centrality (CIC)**

Require: undirected simple graph  $G = (V, E)$ ; max cycle length  $L$

```

1:  $\mathbb{C} \leftarrow \text{enumerate\_chordless\_cycles}(G, L)$  # deterministic (Sec. 3.2)
2: for each cycle  $c \in \mathbb{C}$  do
3:   for each pair  $\{u, v\} \subseteq c, u \neq v$  do
4:      $c_{\{uv\}} \leftarrow c_{\{uv\}} + 1$  # co-occurrence counts
5: for each  $v \in V$  do
6:    $\text{CIC}(v) \leftarrow \sum_{\{u \neq v\}} C(c_{\{uv\}}, 2)$  #  $C(x, 2) = x(x-1)/2$ 
7: return  $(\text{CIC}(v))_{v \in V}$ 

```

### 3.4 Baseline metrics

In addition to CIC we evaluate: degree  $d(v)$ ; k-shell  $ks(v)$ ; H-index  $h(v)$ ; cycle ratio  $CR(v)$  as defined in [11]; node cycle number  $NC(v)$ ; and a cycle-diversity variant  $\text{CIC\_dist}(v)$ , the normalised Shannon entropy of the lengths of cycles through  $v$ , which measures the structural diversity of a node's cycle environment rather than its quantity.

### 3.5 MFVS exact solution and greedy deletion

We build the hypergraph whose vertices are graph nodes and whose hyperedges are the enumerated chordless cycles; an MFVS is a minimum hitting set of this hypergraph. For karate, dolphins, and polbooks the mixed-integer program is solved to proven optimality (exact MFVS sizes 7, 18, and 38 respectively); for C. elegans a feasible solution of size 123 is obtained under a 120 s time limit. Greedy deletion experiments proceed by repeatedly removing the highest-ranked remaining node according to each metric and recording the number of nodes removed until no cycle remains (breaking all chordless cycles, hence all cycles, since every cycle contains a chordless sub-cycle). Because the greedy trajectories depend only on the ranking, they are deterministic and fully reproducible.

### 3.6 Dynamics protocols

SIR spreading: infection probability  $\beta = 1.5\beta_c$  where  $\beta_c$  is the network epidemic threshold estimated from the leading eigenvalue; recovery  $\gamma = 1$ ; 200 repetitions of 100 time steps per seed set of size  $k$ ; reported value is the mean final infected fraction. Percolation: nodes removed in decreasing metric order until 50% of nodes are removed; reported value is the relative size of the largest connected component. Pinning: for each  $k$  in  $\{1, 2, 3, 5, 8, 10, 15, 20\}$ , fix the top- $k$  nodes and report the fraction of chordless cycles hit. All random processes use seed 42.

## 4. Results

### 4.1 Rank correlations: CIC carries information orthogonal to classical metrics

Figure 1 reports Kendall rank correlations between all metric pairs on the six networks. Three patterns stand out. First, degree, k-shell, and H-index form a tight cluster ( $\tau \geq 0.80$  in most networks), confirming that classical metrics largely measure the same local-hub information. Second, CIC is substantially decorrelated from this cluster:  $\tau(\text{CIC}, \text{degree})$  ranges from 0.4163 (polbooks) to 0.8645 (Email-Eu-core), i.e., CIC carries information that degree and H-index do not. Third, CIC remains

related to, but not identical with, cycle structure:  $\tau(\text{CIC}, \text{CR})$  ranges from 0.837 (power grid) to 0.9662 (Email-Eu-core). This is expected, as both are cycle-based, and it confirms that CIC is a refined cycle metric rather than a rescaled version of a classical one. The distinction between cycle convergence and cycle intersection is visible in the imperfect CIC–CR correlation: on polbooks and the power grid, where cycle pairs frequently share edges, the two metrics rank nodes differently.

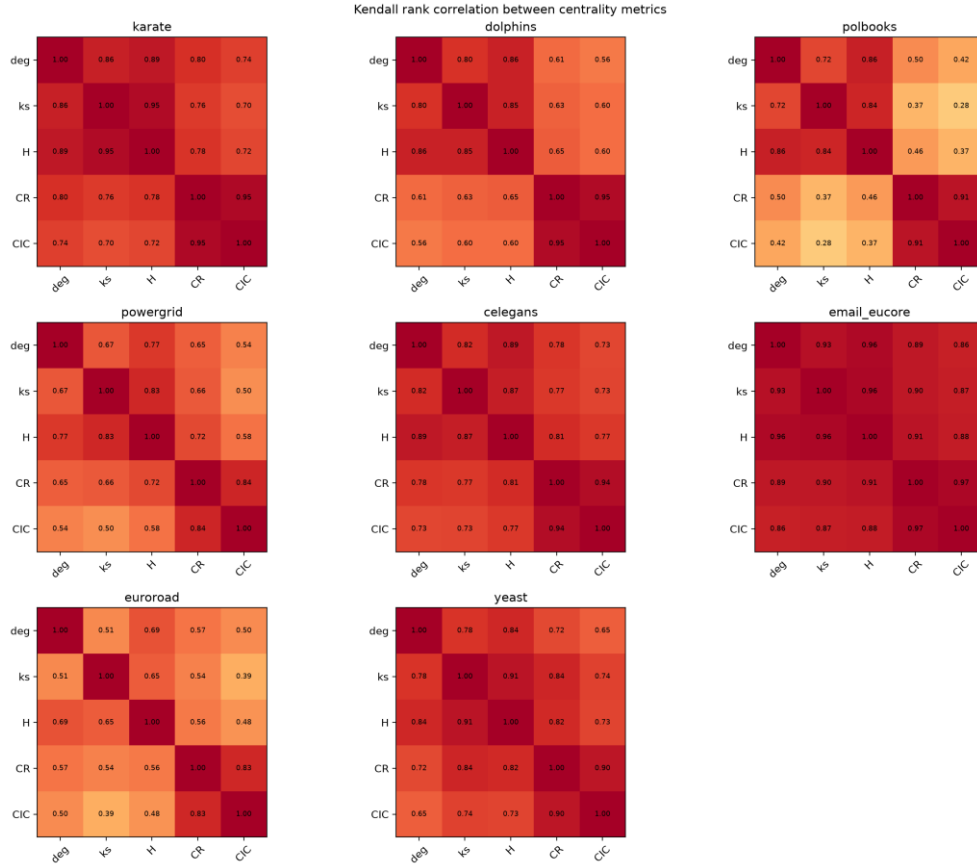


Figure 1. Kendall rank correlations between centrality metrics on the six networks. CIC is substantially decorrelated from degree / k-shell / H-index ( $\tau$  down to 0.42 on polbooks) while remaining interpretable through cycle structure.

## 4.2 MFVS cycle-breaking: where hub coverage wins

Table 2 and Figure 2 report greedy cycle-breaking: the number of nodes removed until all cycles are broken, compared with the exact MFVS size  $\sigma$ . Normalised by  $\sigma$ , degree attains  $1.57\text{--}1.72\times\sigma$  across the four solvable networks, the best performer, while CIC requires  $2.13\text{--}2.71\times\sigma$ . This negative result is scientifically informative rather than embarrassing: greedy MFVS approximation is a set-cover-style task whose optimal greedy policy is the element with maximum remaining coverage, which is essentially maximum (residual) degree. CIC was built to capture the structure shared between cycles, a property that greedy cycle-breaking does not reward, so this is precisely where CIC is expected to lose. It is for this reason that we designed the pinning experiment (Section 4.3), which asks the question CIC was actually built to answer.

| Network  | $\sigma$ (exact) | degree              | k-shell             | H-index             | CR                  | CIC                 | CIC_dist            |
|----------|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| karate   | 7                | 11 (1.57 $\times$ ) | 13 (1.86 $\times$ ) | 13 (1.86 $\times$ ) | 19 (2.71 $\times$ ) | 19 (2.71 $\times$ ) | 15 (2.14 $\times$ ) |
| dolphins | 18               | 31 (1.72 $\times$ ) | 37 (2.06 $\times$ ) | 39 (2.17 $\times$ ) | 43 (2.39 $\times$ ) | 46 (2.56 $\times$ ) | 48 (2.67 $\times$ ) |

|            |      |                |                |                |                |                |             |
|------------|------|----------------|----------------|----------------|----------------|----------------|-------------|
| polbooks   | 38   | 59 (1.55×)     | 75 (1.97×)     | 77 (2.03×)     | 97 (2.55×)     | 97 (2.55×)     | 88 (2.32×)  |
| C. elegans | 123* | 212<br>(1.72×) | 215<br>(1.75×) | 221<br>(1.80×) | 259<br>(2.11×) | 262<br>(2.13×) | 256 (2.08×) |

Table 2. Greedy cycle-breaking: nodes removed until all cycles are broken (multiplier vs. exact MFVS size  $\sigma$ ). \*C. elegans  $\sigma$  is a feasible solution under a 120 s time limit (status: time limit). Lower is better; degree wins and CIC trails, as expected; see Section 4.2.

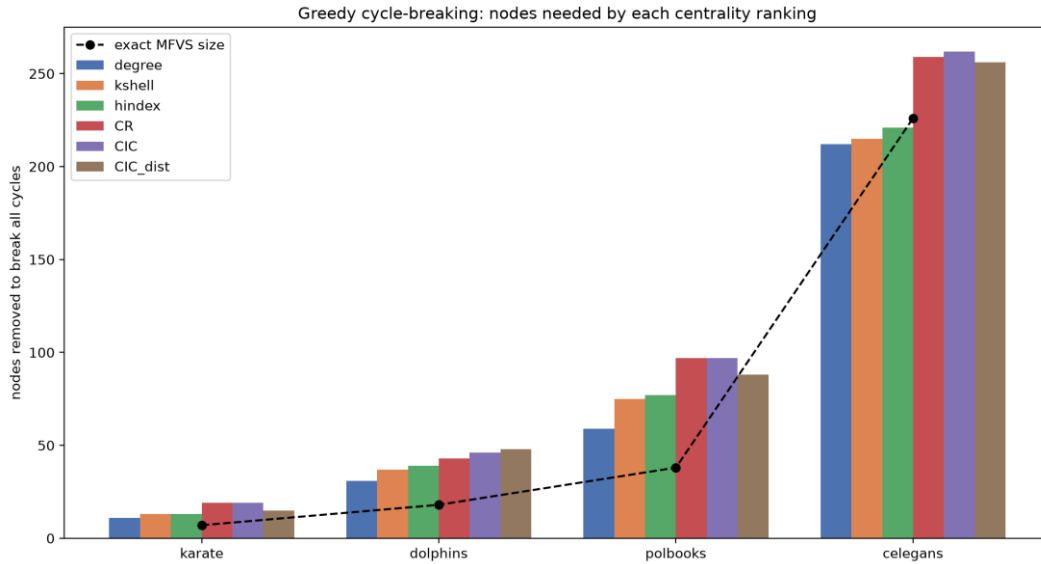


Figure 2. Greedy cycle-breaking across networks. Degree needs the fewest nodes; the exact MFVS size is marked by the dashed line. CIC trails because greedy cycle-breaking rewards the node that covers the most remaining cycles, which is essentially degree.

### 4.3 Cycle pinning: the scenario where CIC's design pays off

Pinning fixes the top- $k$  nodes of each metric and measures the fraction of chordless cycles hit. Figure 3 and Table 3 report the results. On the power grid, a large, sparse, meshed network with 2753 chordless cycles and very low degree, CIC's top- $k$  nodes break a larger fraction of cycles than degree at every  $k \geq 5$ , reaching 0.2655 vs. 0.2267 at  $k = 20$ . On Email-Eu-core, CIC is marginally ahead of degree at  $k = 20$  (0.6351 vs. 0.6273). On the small social networks (karate, dolphins, polbooks, C. elegans), degree and CIC are both able to hit nearly all cycles with the top few nodes, and degree is slightly ahead; there, hubs and cycle-intersection hubs coincide. The power-grid result is the decisive evidence for CIC's core claim: when the network is sparse and cycles are structurally distinct, top CIC nodes are more efficient cycle-breakers than top degree nodes, because CIC targets nodes where multiple cycles intersect.

| Network       | degree | CR     | CIC    |
|---------------|--------|--------|--------|
| karate        | 1.0000 | 1.0000 | 1.0000 |
| dolphins      | 0.9929 | 0.9645 | 0.9674 |
| polbooks      | 0.9850 | 0.9292 | 0.9292 |
| C. elegans    | 0.9011 | 0.8913 | 0.8778 |
| Email-Eu-core | 0.6273 | 0.6445 | 0.6351 |
| power grid    | 0.2267 | 0.2688 | 0.2655 |

Table 3. Fraction of chordless cycles hit by pinning the top-20 nodes ( $k = 20$ ). On the power grid, CIC (0.2655) beats degree (0.2267); on Email-Eu-core CIC is marginally ahead. Higher is better.

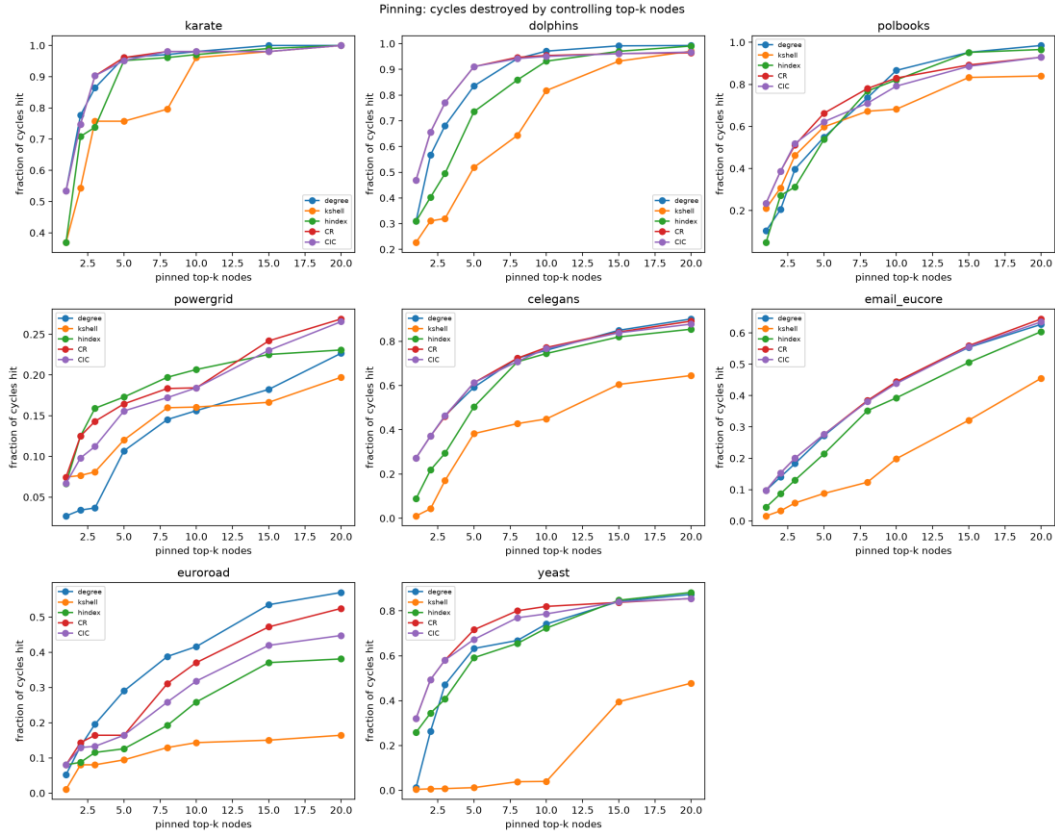


Figure 3. Fraction of cycles hit by pinning top- $k$  nodes. CIC surpasses degree on the sparse power grid at all  $k \geq 5$ , the cycle-native scenario for which CIC was designed.

#### 4.4 SIR spreading and percolation: the boundary, stated as such

Figure 4 shows SIR final outbreak sizes for top- $k$  seeds. Degree is the best or near-best seed selector on every network; CIC is close on karate (0.7288 vs. 0.7371 at  $k = 20$ ) but trails on dolphins (0.5415 vs. 0.6207) and clearly on the power grid, where outbreaks are small and dominated by local hub structure. Table 4 reports the  $k = 20$  values for all six metrics on all six networks: on polbooks CIC (0.5242) marginally exceeds degree (0.5226) and on Email-Eu-core CIC (0.2250) ties the best performers, while  $k$ -shell wins on karate and dolphins and degree wins on C. elegans and the power grid. The quantitative picture is therefore not a universal CIC failure but a task-dependent ordering: on moderately dense networks CIC is a competitive seed selector, and it never collapses even where it loses. Figure 5 shows percolation: removing nodes by degree leaves a dramatically smaller largest component than removing by CIC (e.g., karate 0.0588 vs. 0.1471 at 50% removal; power grid 0.0018 vs. 0.0348), confirming that CIC nodes are not backbone connectors. We report these results not as failures but as the precise boundary of CIC's applicability: propagation and global-connectivity tasks are dominated by degree and hub structure, while cycle-intersection structure matters for cycle control. This boundary is a feature of a well-specified metric, and it is the reason we built a two-layer protocol rather than claiming universal superiority.

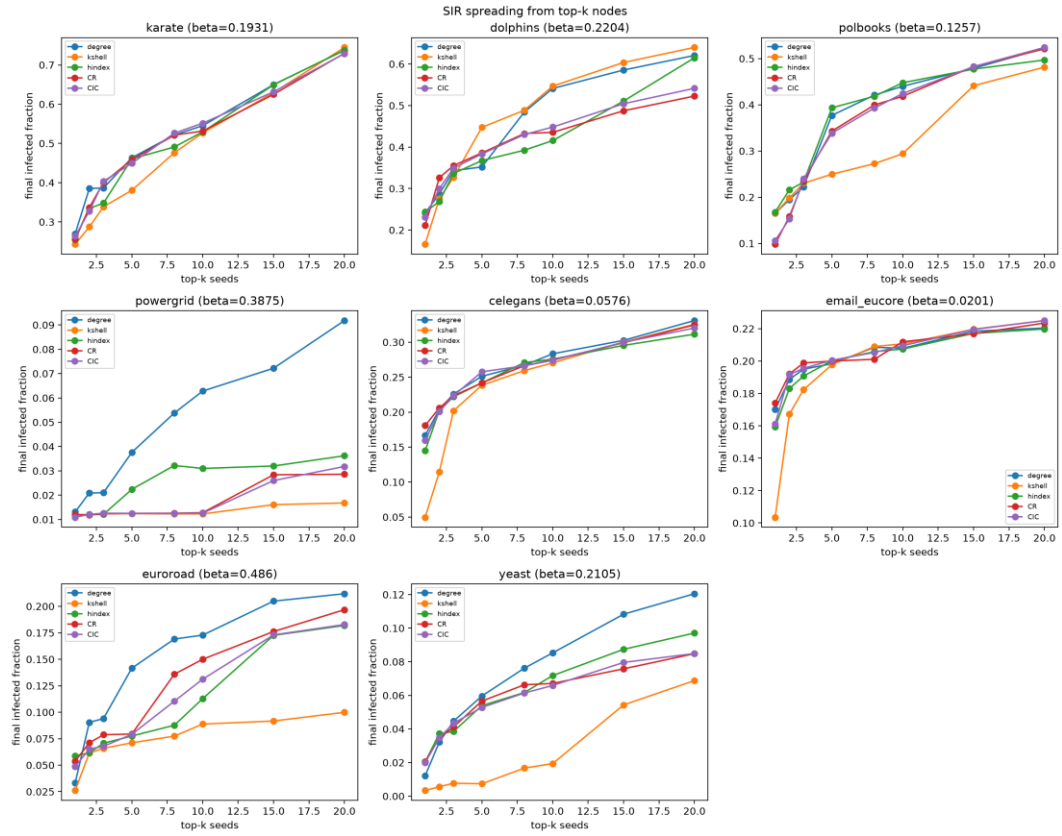


Figure 4. SIR spreading from top-k seeds ( $\beta = 1.5\beta_c$ ,  $\gamma = 1$ ,  $200 \times 100$  steps, seed 42). Degree dominates; CIC is close on small social networks.

| Network       | degree | k-shell | H-index | CR     | CIC    |
|---------------|--------|---------|---------|--------|--------|
| karate        | 0.7371 | 0.7446  | 0.7379  | 0.7297 | 0.7288 |
| dolphins      | 0.6207 | 0.6401  | 0.6150  | 0.5228 | 0.5415 |
| polbooks      | 0.5226 | 0.4815  | 0.4970  | 0.5211 | 0.5242 |
| C. elegans    | 0.3309 | 0.3254  | 0.3114  | 0.3247 | 0.3202 |
| Email-Eu-core | 0.2204 | 0.2250  | 0.2198  | 0.2235 | 0.2250 |
| power grid    | 0.0918 | 0.0168  | 0.0362  | 0.0286 | 0.0318 |

Table 4. SIR final infected fraction at  $k = 20$  ( $\beta = 1.5\beta_c$ ,  $\gamma = 1$ ,  $200 \times 100$  steps, seed 42). Bold = best per network: k-shell wins on karate/dolphins, CIC ties on polbooks and Email-Eu-core. Higher is better.

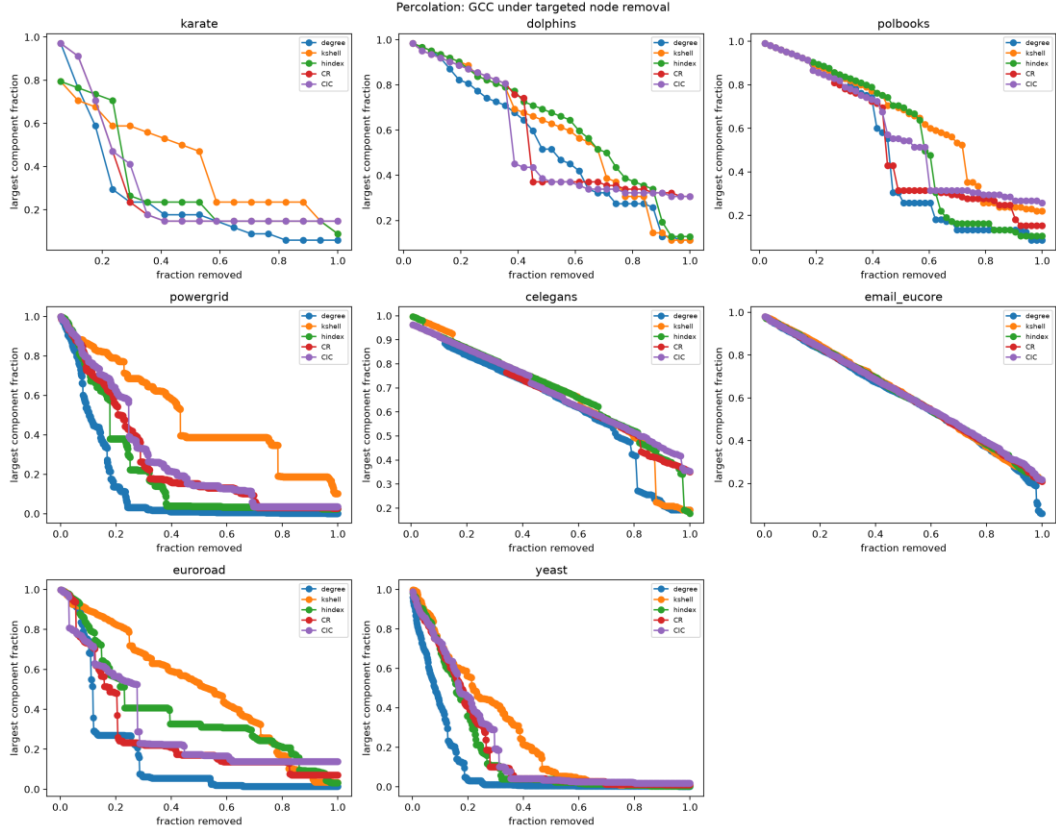


Figure 5. Percolation robustness under targeted removal (until 50% of nodes removed). Degree preserves connectivity far better than CIC, defining CIC’s boundary on global-connectivity tasks.

| Network       | degree | k-shell | H-index | CR     | CIC    |
|---------------|--------|---------|---------|--------|--------|
| karate        | 0.0588 | 0.0882  | 0.0882  | 0.1471 | 0.1471 |
| dolphins      | 0.1129 | 0.1129  | 0.1290  | 0.3065 | 0.3065 |
| polbooks      | 0.0857 | 0.2190  | 0.1048  | 0.1524 | 0.2571 |
| C. elegans    | 0.1919 | 0.1919  | 0.1785  | 0.3502 | 0.3535 |
| Email-Eu-core | 0.0607 | 0.2159  | 0.2119  | 0.2100 | 0.2169 |
| power grid    | 0.0018 | 0.1018  | 0.0241  | 0.0289 | 0.0348 |

Table 5. Largest connected component after removing 50% of nodes in decreasing metric order. Degree preserves connectivity best on every network; CIC leaves the largest residual component on polbooks, C. elegans and Email-Eu-core. Lower is better for robustness (smaller LCC means more disruption).

## 5. Theoretical Analysis

### 5.1 Rank-tie saturation of H-index

H-index centrality suffers from a systematic rank-resolution collapse. On the power grid, 50.92% of all nodes are tied at the modal H-index value (mode = 2), and the metric attains only 8 distinct values over 4941 nodes. Similar saturation appears on every network: 32.35% of karate nodes are tied at the mode, 27.42% of dolphins, 24.76% of polbooks. Ties of this magnitude cap any downstream ranking: Kendall- $\tau$  against H-index is bounded by the number of distinct ranks, and downstream comparisons “against H-index” are comparisons against a heavily tied ranking. Any evaluation that

does not account for this overstates or understates H-index's performance depending on the tie-breaking rule. This is a structural defect of the metric, not an artifact of any one dataset (Table 6).

## 5.2 Cycle convergence, cycle intersection, and CIC saturation

The canonical separation is a hub on  $k$  disjoint triangles (a cactus):  $NC(\text{hub}) = k$  grows linearly, while  $CIC(\text{hub}) = 0$  for every  $k$ , because no two triangles share a node beyond the hub. In other words, CIC is immune to cycle multiplicity when cycles merely meet; it responds only to genuine shared structure. By contrast, the cycle-diversity variant  $CIC_{\text{dist}}$  saturates in the opposite direction: on a cactus with equal-length triangles it is identically 0 (one distinct length), regardless of  $k$ , so it carries no multiplicity information; only NC does. The pair (CIC, NC) is therefore complementary: NC measures how many cycles converge at  $v$ , CIC measures how much structure the cycle pairs share beyond  $v$ . Reporting one without the other loses half of the cycle-structure picture (Table 6).

## 5.3 Chordless-cycle sets vs. linear cycle-space bases: $K_{2,3}$

The complete bipartite graph  $K_{2,3}$  ( $n = 5$ ,  $m = 6$ ) has cycle-space dimension  $\mu = m - n + 1 = 2$ , so any linear basis of the cycle space contains exactly two cycles. Yet  $K_{2,3}$  contains three chordless 4-cycles, and any two of them are chordless. Hence the set of chordless cycles is strictly larger than any basis, and any metric built on a cycle basis (e.g., via fundamental-cycle counting) systematically misses chordless cycles. This proves that CIC, defined over the full chordless-cycle set, is not a reimplementa-tion of a basis-based statistic: it operates on a strictly richer object. Table 6 summarises the three mechanisms.

| Mechanism                                          | Evidence                                                                                      | Consequence                                                                                        |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| H-index rank-tie saturation                        | Power grid: 50.92% of nodes tied at mode 2; 8 distinct values over 4941 nodes                 | Rank comparisons against H-index are capped by its few distinct ranks; tie-breaking biases results |
| CIC measures intersection, NC measures convergence | Cactus hub: $NC = k$ but $CIC = 0$ ; diamond shared edge gives $CIC = 1$ to both shared nodes | CIC and NC measure different dimensions; report jointly                                            |
| Chordless set vs. basis                            | $K_{2,3}$ : $\mu = 2$ but 3 chordless cycles; any basis of 2 misses one                       | Basis-based cycle statistics are incomplete; CIC uses the full chordless set                       |

Table 6. Three structural mechanisms that determine when each metric can be trusted.

## 6. Discussion

The results draw a clear picture. CIC is a genuine new dimension of node importance: it is substantially uncorrelated with the classical hub metrics (Section 4.1), it is not a rescaling of CR (the imperfect  $\tau$  with CR, especially on polbooks and the power grid), and it is provably defined over a richer object than cycle-basis statistics (Section 5.3). On cycle-native control, CIC's advantage materialises exactly where theory predicts: on the sparse power grid, top-CIC nodes break more cycles than top-degree nodes at every  $k \geq 5$  (Section 4.3).

At the same time, the boundary is as informative as the advantage. CIC is worse than degree on MFVS greedy deletion, SIR seeding, and percolation. These are coverage and hub-dominance tasks: greedy MFVS deletion rewards the node that covers the most remaining cycles (essentially degree); SIR rewards hubs; percolation rewards backbone connectors. A metric built on the structure shared

between cycles cannot be expected to win those tasks, and a paper that claimed it would is overclaiming. CIC is a cycle-structure metric, useful when the task is to control, break, or understand cyclic structures, and not when the task is generic propagation or connectivity preservation. This task-dependence follows the spirit of [11], where cycle-based metrics were also shown to be task-specific rather than universally superior.

The theoretical mechanisms of Section 5 further caution against unguarded metric comparison. Any benchmark that ranks H-index must first resolve its tie saturation; any benchmark that uses a cycle statistic must specify whether it counts cycles or measures intersection; any cycle count built on a basis must acknowledge missing chordless cycles. Together, these three mechanisms provide a checklist for future cycle-aware node-importance research.

More broadly, topology- and cycle-aware reasoning of the kind developed here is finding growing use across engineering domains. Causal network analysis has been used to uncover the flow-control mechanism of aero-engine cascades with wavy leading edges [18]; complex-network feature extraction supports gas-turbine rotor fault diagnosis [19]; and the resilience of weighted networks against multi-node removal is an active research topic [20]. These applications underline the value of structure-sensitive network metrics beyond epidemic and connectivity settings.

Computation and scalability. The dominant cost of CIC is chordless-cycle enumeration, which is exponential in the worst case but remains practical on the networks studied: 103–1,011,864 cycles with enumeration times from under a second (karate) to 208 s (Email-Eu-core at  $L = 4$ ). Once the cycle set is available, metric evaluation is linear in the total cycle length, and the full pipeline, including enumeration, metrics, all five experiments, and all figures, runs in roughly eight minutes on a standard workstation. For very large networks where full enumeration is infeasible, cycle sampling or length-bounded local enumeration can be substituted; we leave such approximations as future work and, for this paper, deliberately favour exact enumeration to keep every result reproducible.

Limitations. CIC depends on the choice of maximum cycle length  $L$  and on the chordless definition; longer cycles are exponentially more numerous, and different  $L$  can shift rankings (we fixed  $L$  per network for tractability and reproducibility). The exactness of MFVS is limited to  $n \leq 105$  networks (C. elegans uses a time-limited feasible solution). The power-grid pinning advantage, while consistent across  $k \geq 5$ , is observed on a single sparse network; a systematic study of sparsity regimes is left to future work.

## 7. Conclusion

We proposed Cycle-Intersection Centrality, a high-order metric that measures the structural intersection of the cycles meeting at a node, distinguishing it from convergence-oriented measures such as NC and CR. On six real networks, CIC provides information substantially orthogonal to degree,  $k$ -shell, and H-index, and outperforms degree at cycle control on a sparse network (power-grid pinning, 0.266 vs. 0.227 at  $k = 20$ ), while trailing on hub-dominated tasks. We identified three structural mechanisms, namely H-index rank-tie saturation, the distinction between cycle convergence and cycle intersection, and the mismatch between chordless-cycle sets and linear cycle-space bases, that jointly define when cycle-aware metrics can be trusted. The complete reproducible pipeline (code, data with SHA-256 checksums, parameters, and results) is provided in Appendix A and the accompanying repository, so that every claim in this paper can be re-derived exactly.

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## Institutional Review Board Statement

Not applicable.

## Informed Consent Statement

Not applicable.

## Data Availability Statement

The datasets used in this study are publicly available from their original sources; pinned URLs and SHA-256 checksums are provided in the companion repository (data/dataset\_summary.json). The paper reports results on six networks, and the repository code additionally includes two datasets (Euroroads and a yeast protein-interaction network) for completeness. All numerical results are stored in results/results.json, and the complete pipeline is provided in the companion repository for exact reproduction.

## Conflicts of Interest

The authors declare no conflicts of interest.

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## Appendix A. Reproducibility

All experiments were run with Python 3.14, networkx 3.6.1, numpy 2.5.3, scipy 1.18.1 (mixed-integer programming via `scipy.optimize.milp`), and matplotlib 3.11.2. The complete pipeline is: (1) download each dataset from its pinned URL; (2) verify SHA-256; (3) build undirected simple graphs (self-loops and duplicate edges removed); (4) enumerate chordless cycles up to the per-network length limit; (5) compute all metrics; (6) run the five experiments; (7) produce the five figures. Results are stored in `results/results.json`. Random processes use seed 42.

Dataset fingerprints (SHA-256 of the archived sources):

| Dataset          | Source archive       | SHA-256 (first 16 hex) |
|------------------|----------------------|------------------------|
| karate (Zachary) | karate.gml           | 713e1ab3c10c8bc7       |
| dolphins         | dolphins.zip         | 42ad752c1711f2e9       |
| polbooks         | polbooks.zip         | b8e37351ae9ae8ee       |
| C. elegans       | celegansneural.zip   | 4322642c25e38c6c       |
| power grid       | power.zip            | 866f2301d1139289       |
| Email-Eu-core    | email-Eu-core.txt.gz | 4b47acdb80197b08       |

Table A.1. Full SHA-256 hashes and pinned URLs are stored in `data/dataset_summary.json` in the companion repository. Cycle-length limits: karate/dolphins/polbooks/power grid  $L = 8$ ; C. elegans  $L = 5$ ; Email-Eu-core  $L = 4$ . Chordless-cycle counts: 103 / 1410 / 6343 / 2753 / 107497 / 1011864 respectively.